12. HAFTA

Hatırlatma

* Eksponsiyonel dereceli ve her sonlu $0 \leq x \leq b$ aralığında parçalı sürekli f ve g fonksiyonlarının konvolusyonu

$$f(x) * g(x) = \int_0^x f(z)\, g(x-z)\, dz$$

integrali ile belirlenmiş f*g fonksiyonuna denir.

Ayrıca, $f*g = g*f$ dir

* Eğer f ve g fonksiyonları e^{at} eksponsiyonel dereceli ve her $0 \leq x \leq b$ aralığında parçalı sürekli iki fonksiyon olursa, o halde f*g konvolusyonun Laplace dönüşümü için

$$L[f*g] = L[f] \cdot L[g]$$

formülü geçerlidir.

Böylece $L^{-1}[F(s) \cdot G(s)] = L^{-1}[F(s)] * L^{-1}[G(s)]$ elde edilir.

Alıştırmalar

1) $H(s) = \dfrac{1}{s^2+5s+6}$ fonksiyonu için $L^{-1}[H(s)]$ yi

(a) Basit kesirlere ayırarak

(b) Konvolusyon kullanarak

çözünüz.

cvp: (a) $H(s) = \dfrac{1}{s^2+5s+6} = \dfrac{1}{(s+3)(s+2)} = \dfrac{A}{s+3} + \dfrac{B}{s+2} \Rightarrow A=-1$ ve $B=1$

$$\Rightarrow H(s) = \frac{-1}{s+3} + \frac{1}{s+2} \Rightarrow L^{-1}[H(s)] = -L^{-1}\left[\frac{1}{s+3}\right] + L^{-1}\left[\frac{1}{s+2}\right] \Rightarrow$$

$$\Rightarrow \mathcal{L}^{-1}[H(s)] = -e^{-3x} + e^{-2x}$$

(b) $H(s) = \dfrac{1}{s^2+5s+6} = \dfrac{1}{(s+3)(s+2)} = \dfrac{1}{s+3} \cdot \dfrac{1}{s+2}$

$$\Rightarrow \mathcal{L}^{-1}[H(s)] = \mathcal{L}^{-1}\left(\frac{1}{s+3}\right) * \mathcal{L}^{-1}\left(\frac{1}{s+2}\right) \Rightarrow \mathcal{L}^{-1}[H(s)] = e^{-3x} * e^{-2x}$$

$$\Rightarrow \mathcal{L}^{-1}[H(s)] = \int_0^x e^{-3\tau} e^{-2(x-\tau)} d\tau = e^{-2x} \int_0^x e^{-\tau} d\tau = -e^{-2x} e^{-\tau}\Big|_0^x = \underline{-e^{-3x} + e^{-2x}}$$

2) $H(s) = \dfrac{1}{s(s^2+9)}$ fonksiyonu için $\mathcal{L}^{-1}[H(s)]$ yi

(a) Basit kesirlere ayırarak

(b) Konvolusyon kullanarak

Çözünüz.

cvp: (a) $H(s) = \dfrac{1}{s(s^2+9)} = \dfrac{A}{s} + \dfrac{Bs+C}{s^2+9} \Rightarrow 1 = A(s^2+9) + (Bs+C)s$

(s^2+9) (s)

$$\Rightarrow A = \frac{1}{9}, \quad B = -\frac{1}{9}, \quad C = 0$$

$$\Rightarrow H(s) = \frac{1}{9} \cdot \frac{1}{s} - \frac{1}{9} \frac{s}{s^2+9} \Rightarrow \mathcal{L}^{-1}[H(s)] = \frac{1}{9}\mathcal{L}^{-1}\left[\frac{1}{s}\right] - \frac{1}{9}\mathcal{L}^{-1}\left[\frac{s}{s^2+3^2}\right]$$

$$\Rightarrow \mathcal{L}^{-1}[H(s)] = \frac{1}{9} - \frac{\cos 3x}{9} = \frac{1-\cos 3x}{9}$$

(b) $H(s) = \dfrac{1}{s(s^2+9)} = \dfrac{1}{s} \cdot \dfrac{1}{s^2+9} = \dfrac{1}{s} \cdot \dfrac{1}{3} \dfrac{3}{s^2+9}$

$\Rightarrow \mathcal{L}^{-1}[H(s)] = \mathcal{L}^{-1}\left[\dfrac{1}{s}\right] * \mathcal{L}^{-1}\left[\dfrac{1}{3} \dfrac{3}{s^2+3^2}\right] \Rightarrow \mathcal{L}^{-1}[H(s)] = 1 * \dfrac{\sin 3x}{3}$

$\Rightarrow \mathcal{L}^{-1}[H(s)] = \displaystyle\int_0^x \frac{\sin 3\tau}{3}\, d\tau = -\frac{1}{3}\frac{\cos 3\tau}{3}\Big|_0^x = \frac{1-\cos 3x}{9}$

3) $\dfrac{dy}{dx} - y = e^{3x}$, $y(0)=2$ başlangıç-değer problemini Laplace dönüşümü uygulayarak çözünüz

cvp: $y' - y = e^{3x} \Rightarrow \mathcal{L}[y'] - \mathcal{L}[y] = \mathcal{L}[e^{3x}]$

$\Rightarrow s\mathcal{L}[y] - y(0) - \mathcal{L}[y] = \dfrac{1}{s-3} \underset{(y(0)=2)}{\Longrightarrow} \mathcal{L}[y](s-1) = \dfrac{1}{s-3} + 2$

$\Rightarrow \mathcal{L}[y] = \dfrac{1}{(s-3)(s-1)} + \dfrac{2}{s-1} \Rightarrow \mathcal{L}[y] = \dfrac{1}{2}\dfrac{1}{s-3} - \dfrac{1}{2}\dfrac{1}{s-1} + \dfrac{2}{s-1}$

$\downarrow$

$\left[\dfrac{1}{(s-3)(s-1)} = \dfrac{A}{s-3} + \dfrac{B}{s-1} \Rightarrow A = \dfrac{1}{2}, B = -\dfrac{1}{2}\right]$

$\Rightarrow y(x) = \dfrac{1}{2}\mathcal{L}^{-1}\left[\dfrac{1}{s-3}\right] - \dfrac{1}{2}\mathcal{L}^{-1}\left[\dfrac{1}{s-1}\right] + 2\mathcal{L}^{-1}\left[\dfrac{1}{s-1}\right]$

$\Rightarrow y(x) = \dfrac{e^{3x}}{2} - \dfrac{e^x}{2} + 2e^x \Rightarrow \boxed{y(x) = \dfrac{3e^x + e^{3x}}{2}}$ //

4) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 18e^{-x}\sin 3x$; $y(0)=0$, $y'(0)=3$

başlangıç-değer problemini Laplace dönüşümü kullanarak çözünüz.

cvp: $y'' - y' - 2y = 18e^{-x}\sin 3x \Rightarrow L[y''] - L[y'] - 2L[y] = 18L[e^{-x}\sin 3x]$

$\Rightarrow s^2L[y] - \cancel{sy(0)}^{0} - y'(0) - sL[y] + \cancel{y(0)}^{0} - 2L[y] = 18L[e^{-x}\sin 3x]$

$\Rightarrow L[y](s^2-s-2) = \frac{18\cdot 3}{(s+1)^2+9} + 3 \Rightarrow L[y] = \frac{54}{(s^2+2s+10)(s^2-s-2)} + \frac{3}{s^2-s-2}$

$$\# \quad \frac{54}{(s^2+2s+10)(s-2)(s+1)} = \frac{A}{s-2} + \frac{B}{s+1} + \frac{Cs+D}{s^2+2s+10}$$

$(s^2+2s+10)(s+1)$ $(s^2+2s+10)(s-2)$ $(s-2)(s+1)$

$\Rightarrow 54 = A(s^2+2s+10)(s+1) + B(s^2+2s+10)(s-2) + (Cs+D)(s-2)(s+1)$

$B=-2$, $A=1$, $C=1$, $D=-2$

$$\# \quad \frac{3}{(s-2)(s+1)} = \frac{E}{s-2}_{(s+1)} + \frac{F}{s+1}_{(s-2)} \Rightarrow 3 = E(s+1) + F(s-2) \Rightarrow E=1,\ F=-1$$

$$\Rightarrow y(x) = L^{-1}\left[\frac{54}{(s^2+2s+10)(s-2)(s+1)}\right] + L^{-1}\left[\frac{3}{s^2-s-2}\right]$$

$$= L^{-1}\left[\frac{1}{s-2}\right] - 2L^{-1}\left[\frac{1}{s+1}\right] + L^{-1}\left[\frac{s-2}{s^2+2s+10}\right] + L^{-1}\left[\frac{1}{s-2}\right] - L^{-1}\left[\frac{1}{s+1}\right]$$

$$= e^{2x} - 2e^{-x} + e^{2x} - e^{-x} + L^{-1}\left[\frac{s+1-3}{(s+1)^2+3^2}\right]$$

$$\Rightarrow y(x) = 2e^{2x} - 3e^{-x} + L^{-1}\left[\frac{s+1}{(s+1)^2+3^2}\right] - L^{-1}\left[\frac{3}{(s+1)^2+3^2}\right]$$

$$\Rightarrow \boxed{y(x) = 2e^{2x} - 3e^{-x} + e^{-x}\cos 3x - e^{-x}\sin 3x}$$

5) $\begin{cases} \dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = h(x) \\ y(0) = 0 \ , \ y'(0) = 0 \end{cases}$, $h(x) = \begin{cases} 2 \ , & 0 < x < 4 \\ 0 \ , & x > 4 \end{cases}$

başlangıç-değer problemini Laplace dönüşümü uygulayarak çözünüz.

cvp: Öncelikle $h(x)$ fonksiyonunun Laplace dönüşümünü hesaplayalım.

$$L[h(x)] = \int_0^\infty e^{-sx} h(x)\,dx = 2\int_0^4 e^{-sx}\,dx = \frac{-2}{s} e^{-sx}\Big|_0^4 = -\frac{2e^{-4s}}{s} + \frac{2}{s}$$

$$y'' - 3y' + 2y = h(x) \Rightarrow L[y''] - 3L[y'] + 2L[y] = L[h(x)]$$

$$\Rightarrow s^2L[y] - \cancel{sy(0)} - \cancel{y'(0)} - 3sL[y] + \cancel{3y(0)} + 2L[y] = L[h(x)]$$

$$\Rightarrow L[y](s^2 - 3s + 2) = \frac{2}{s}(1 - e^{-4s}) \Rightarrow L[y] = \frac{2}{s(s-2)(s-1)}(1 - e^{-4s})$$

$$\Rightarrow L[y] = \frac{2}{s(s-2)(s-1)} - \frac{2e^{-4s}}{s(s-2)(s-1)} \Rightarrow y(x) = L^{-1}\left[\frac{2}{s(s-2)(s-1)}\right] - L^{-1}\left[\frac{2e^{-4s}}{s(s-2)(s-1)}\right]$$

$\frac{2}{s(s-2)(s-1)} = \frac{A}{s} + \frac{B}{s-2} + \frac{C}{s-1} \Rightarrow 2 = A(s-2)(s-1) + Bs(s-1) + Cs(s-2)$

$\frac{A}{(s-2)(s-1)}$ $\frac{B}{s(s-1)}$ $\frac{C}{s(s-2)}$

$\Rightarrow A=1 \;,\; B=1 \;,\; C=-2$

$$\Rightarrow \mathcal{L}^{-1}\left[\frac{2}{s(s-2)(s-1)}\right] = \mathcal{L}^{-1}\left[\frac{1}{s}\right] + \mathcal{L}^{-1}\left[\frac{1}{s-2}\right] - 2\mathcal{L}^{-1}\left[\frac{1}{s-1}\right]$$

$$= 1 + e^{2x} - 2e^{x}$$

$f(x) = \begin{cases} 0 & 0<x<4 \\ g(x-4) & x>4 \end{cases}$ o.ü., $f(x) = u_4(x)\cdot g(x-a)$ olduğunu ve

bu sebepten $\mathcal{L}[f(x)] = \mathcal{L}[u_4(x)\cdot g(x-a)] = e^{-4s}\mathcal{L}[g(x)]$ olduğunu biliyoruz

o halde, $\mathcal{L}[g(x)] = \frac{2}{s(s-2)(s-1)}$ olur. Buradan, $g(x) = 1+e^{2x}-2e^{x}$ elde edilir.

$$\mathcal{L}[f(x)] = e^{-4s}\mathcal{L}[g(x)] \Rightarrow \mathcal{L}[f(x)] = e^{-4s}\cdot\frac{2}{s(s-2)(s-1)} \Rightarrow \mathcal{L}^{-1}\left[\frac{2e^{-4s}}{s(s-2)(s-1)}\right] = f(x)$$

$$\Rightarrow \mathcal{L}^{-1}\left[\frac{2e^{-4s}}{s(s-2)(s-1)}\right] = \begin{cases} 0 & , \; 0<x<4 \\ 1+e^{2(x-4)} - 2e^{x-4} & , \; x>4 \end{cases}$$ bulunur.

Bu durumda,

$$y(x) = \begin{cases} 1+e^{2x}-2e^{x} & 0<x<4 \\ (1-e^{-8})e^{2x} + 2(e^{-4}-1)e^{x} & x>4 \end{cases}$$ elde edilir.